

Internal Product Development Department and Outsourcing Team Follow-up Meeting

Participants:

- Diana Müller (Director of Product Development)
- Markus Weber (Director of Technology Integration)
- Sophia Schmidt (Lead Developer)
- Carlos Herrera (Product Owner - Customer Tracking)
- Hannah Fischer (Project Manager)

Outsourcing Representatives:

- John Smith (Delivery Manager)
- Emma Lee (Business Analyst)
- Ravi Patel (Solution Architect)
- Maria Gonzalez (UX/UI Designer)

[Virtual Meeting - Conference Room, 10:00 AM CET]

Diana (Director of Product Development): Good morning, everyone. Thanks for coming to this follow-up meeting. Today, we want to finalize the priorities and set a clear path forward so the outsourcing team can start their work. We need to be specific about what should be done first, and what comes next, considering all the discussions we've had so far. John, do you want to kick us off?

John (Delivery Manager): Thanks, Diana. To recap, we're dealing with several major issues: data synchronization delays, inconsistent data transformation, outdated infrastructure, notification delays, and UX improvements. I think we should start by focusing on the **foundational issues**, specifically data synchronization and infrastructure, before moving on to user-facing features like notifications and UI improvements. That way, we can solve the root causes of many of the user complaints. What are your thoughts?

Markus (Director of Technology Integration): I agree. Our current infrastructure is holding us back. We need to stabilize the backend first so we can support more dynamic front-end features later. I think the first priority should be implementing **event-driven data synchronization**. Ravi, you mentioned using Kafka or another message broker. Could you elaborate on how we might get started with that?

Ravi (Solution Architect): Absolutely. The first step would be to introduce a **message broker** like Kafka to manage updates between the WMS, OMS, and the portal. We can set up producers in the WMS and OMS to push changes to a Kafka topic, and consumers that update the portal's cache in real-time. This will significantly improve data freshness and reduce the load on manual synchronization processes.

Sophia (Lead Developer): That sounds like a good approach. We'll need to map out where we'll add producers and consumers. The on-prem systems are a bit outdated, so we'll need to assess how feasible it is to integrate Kafka with those systems. It might require some middleware changes too.

Emma (Business Analyst): That makes sense. I think we also need to clearly define the **events** that should trigger updates—things like shipment status changes, estimated delivery times, and any exceptions. We need to ensure that we capture all critical events to keep data current across all systems.

Diana (Director of Product Development): Agreed. So, let's mark **event-driven synchronization** as our first priority. Next, I think we should tackle the **data transformation issues**. Carlos, you mentioned that the current middleware is a black box, and that's causing problems. How should we approach fixing this?

Carlos (Product Owner - Customer Tracking): I think we need to break the middleware down into more **modular services**. Right now, if something goes wrong, it's almost impossible to trace back the source of the error. If we could separate the different

transformation tasks—like unit conversion, timestamp normalization, and code mapping—it would make debugging much easier and improve reliability.

Ravi (Solution Architect): We could use a **microservices architecture** for this. Each transformation task could be handled by a dedicated microservice, allowing us to isolate failures and make it easier to introduce new transformations in the future. We can containerize these services and deploy them in the cloud to make scaling simpler.

Hannah (Project Manager): Okay, so the **second priority** is to break down the middleware into modular transformation services using a microservices approach. Once we have that, we can more easily diagnose and fix data issues.

John (Delivery Manager): Great. Now, for the **third priority**, I think we should work on improving **notification systems**. From what I gathered, the main issue is the delay between triggering events and the notifications going out. If we address the first two priorities, the notifications should automatically become more timely, but we also need to make sure the notification system is capable of real-time processing.

Markus (Director of Technology Integration): Yes, we need to update the notification system to be **event-driven** as well. Once we have Kafka in place, we could use it to trigger notifications instantly rather than relying on batch processes. We also need to add support for **multi-channel notifications**—SMS, push, and email—to give customers more options.

Maria (UX/UI Designer): I'd like to add that we should also think about **designing new notification templates** that are clearer and more informative. Right now, users often get vague messages that don't help them understand what's happening with their shipment. Adding more context to the notifications would improve the customer experience.

Emma (Business Analyst): So, to summarize the **third priority**: make the notification system event-driven, add multi-channel capabilities, and redesign the notification templates to be more user-friendly.

Diana (Director of Product Development): That makes sense. Once we've tackled the backend improvements and notifications, I think we can start working on **user interface improvements** as our **fourth priority**. Maria, can you speak to what you think is most urgent on the UI side?

Maria (UX/UI Designer): Definitely. The **loading times** are the biggest issue right now. Once we have the improved backend and event-driven architecture in place, we can add **loading indicators** and **caching strategies** to make the UI feel more responsive. We also need to update the **shipment tracking pages** to make them clearer for users—adding features like expected delivery windows, which will be more accurate once the backend is improved.

John (Delivery Manager): Perfect. So just to recap our **priorities**:

1. Implement **event-driven data synchronization** using Kafka to improve data freshness and reduce manual processes.
2. Refactor the **data transformation middleware** into modular microservices to improve traceability and reliability.
3. Improve the **notification system** by making it event-driven, adding multi-channel support, and redesigning the templates.
4. Enhance the **user interface** with better loading indicators, caching strategies, and clearer shipment tracking features.

Diana (Director of Product Development): That sounds like a solid plan. Let's focus on getting the first priority underway. Markus and Sophia, I'd like you to work with Ravi and Emma to create a detailed integration plan for Kafka. We need to identify all the touchpoints in our current systems and any potential blockers.

Markus (Director of Technology Integration): Absolutely. We'll get started on that and outline a phased integration approach.

Sophia (Lead Developer): I'll also start assessing the existing on-prem systems to determine what modifications are needed for Kafka integration.

John (Delivery Manager): Great. We'll draft a project timeline based on these priorities and share it with you by the end of the week. We want to make sure everyone is aligned before we begin implementation.

Diana (Director of Product Development): Perfect. Thanks, everyone, for your input. Let's make this a priority and meet again next week to check progress. If anything comes up in the meantime, feel free to reach out.

[End of Meeting]